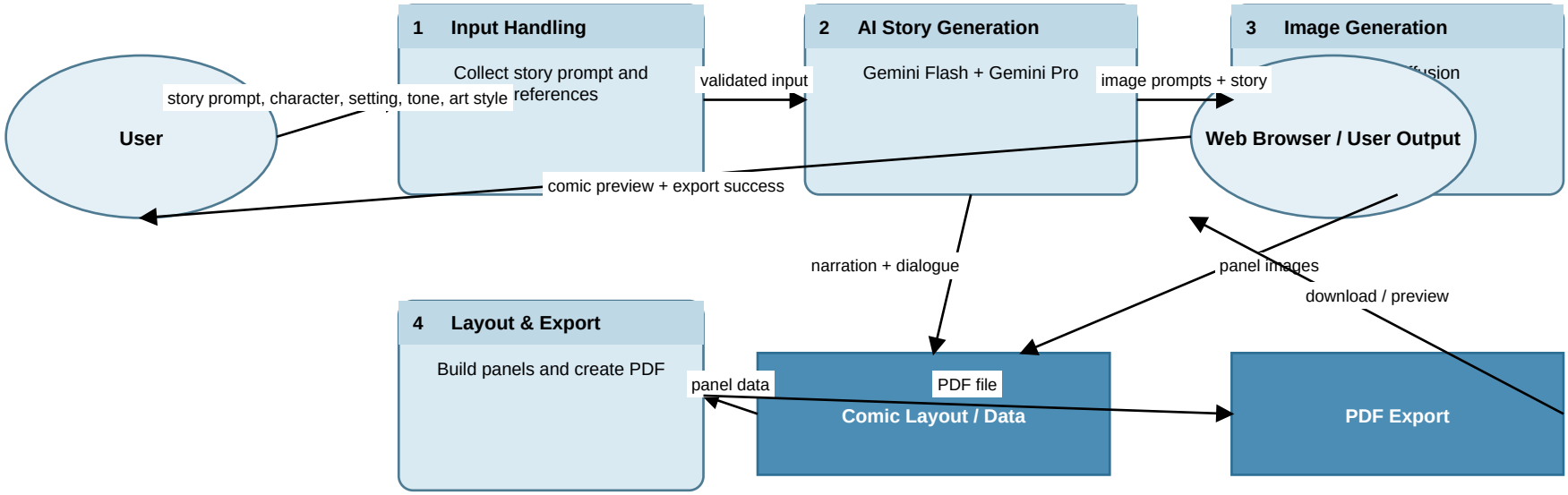
Data Flow Diagram

Date	15 March 2026	**Team ID**	SWTID-2026-3296
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models	**Team Size**	4



ComicCraft flow: FastAPI receives input -> Gemini models generate outline/story -> Stable Diffusion generates images -> layout builder organizes panels -> PDF exporter creates final comic.

DFD Data Flow Details - ComicCraft

Component	Role in the system	Main data handled
External Entity - User	Provides the creative input and receives the generated comic / PDF.	Story prompt, character name, setting, tone, art style; preview and exported PDF.
1. Input Handling	FastAPI receives and validates form or JSON input.	Validated user inputs.
2. AI Story Generation	Gemini Flash creates the structured panel outline; Gemini Pro creates detailed narration and dialogue.	5-panel outline, narration, dialogues, image prompts.
3. Image Generation	Stable Diffusion generates comic-style illustrations from image prompts.	Panel image prompts and generated images.
4. Layout & Export	Layout builder matches panel images with stories; PDF exporter compiles the final comic.	Panel layout, captions, narration, images, PDF path.
Data Store - Comic Layout / Data	Holds structured panel information used by the layout and export steps.	Panel number, image path, text and layout data.
Data Store - PDF Export	Stores the generated PDF file for download.	Final multi-page comic PDF.

Team Members

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